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RESEARCH ARTICLE

Estimation-based disturbance adaptive model predictive control for wheeled biped robots

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ABSTRACT Enhancing the motion performance of wheeled biped robots amidst uncertain disturbances remains a challenge due to their under-actuated and inherently unstable nature. Aiming to address this issue, this paper proposes a disturbance adaptive control framework for such robots. The framework introduces a disturbance variable to describe the comprehensive effect of disturbances due to environmental interactions on the robotic system. A Kalman filter is also employed to estimate the robot's center of mass (CoM) state and the uncertain disturbances by leveraging the dynamic coupling intrinsic to the robots. Estimated results are then integrated into a nominal model predictive control framework to generate an optimal CoM trajectory over a finite time horizon. This approach enables the robot to adapt to various types of external disturbances in the sagittal plane while maintaining accurate velocity tracking. The efficacy of the proposed approach is validated by conducting experimental evaluations on a hydraulically driven wheeled biped robot.

Keywords wheel-legged robot, adaptive model predictive control, disturbance estimation, Kalman filter

1 Introduction

In recent years, the widespread application of mobile robots has led to increasing demands for their mobility and terrain adaptability. Wheel-legged robots, which combine the fast and energy efficient characteristics of wheeled robots with the agile ambulation capabilities innate to legged robots, have made remarkable progress [1–4]. Among them, wheeled biped robots have been shown to counteract the overturning torque induced by payloads by repositioning their center of mass (CoM), enabling their navigation and manipulation within confined spaces. However, accurately modeling the interactions between robotic systems and physical environments remains challenging. The inherent nonlinear, under-actuated, and non-minimum phase characteristics of wheeled biped robots contribute to their sensitivity to modeling inaccuracies, potentially resulting in velocity tracking errors that pose complications for autonomous planning and navigation. Thus, wheeled biped robots must possess robust adaptability to disturbances arising from environmental interactions. This adaptability is crucial for maintaining motion stability and ensuring accurate velocity tracking, thereby facilitating efficient and precise manipulations.

The uncertain disturbances affecting the robot are mainly attributed to the environmental interaction and inaccurate inertia parameters, which, combined with the dynamic of the nominal CoM, influence the overall dynamic behavior of the robot. In legged robots, the nominal CoM state can be estimated by leveraging joint encoder data alongside kinematic models [5,6]. The environmental interaction forces can be directly measured using force/torque (F/T) sensors installed at specific locations, such as the end-effort of legs or arms, enabling the calculation of ground reaction forces (GRFs) or load forces. However, this method is not only limited to measuring forces at specific locations but is also ineffective against random external disturbances or modeling errors that the robot may encounter. Aiming to improve the versatility of disturbance estimators, researchers have developed various disturbance observers that leverage the inherent dynamic characteristics of the robot,

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including momentum-based observers [7,8], zero-momentum-point-based observers [9], and whole-body-dynamic-based observers [10,11]. Under the premise of known GRF, these methods estimate disturbances by observing the residuals between the actual and nominal states of the robot; they have also been successfully applied to biped and quadrupedal robots. However, such approaches cannot be applied to wheel-legged robots because the wheel ground contact force cannot be directly measured. Therefore, an urgent problem involving comprehending the dynamic characteristics of wheeled biped robots and developing corresponding disturbance estimators must be addressed. The authors believe that current research on disturbance resistance of wheeled biped robots has predominantly remained at a qualitative level, aiming to enhance the motion stability of robots under disturbances from uneven terrains and external forces [12,13]; quantitative analysis of performance metrics such as velocity tracking accuracy remains insufficient. Additionally, research on disturbance estimation for wheeled biped robots is notably limited. Raza et al. [14] analyzed the impact of external disturbances on the motion performance of wheeled biped robots. Xin et al. [15] proposed a method based on uncertain and disturbance estimation using the dynamic of wheeled inverted pendulum (WIP) model, aiming to maintain stable robot motion under time-varying external disturbances. However, further research is necessary to enhance the robot's capability to estimate and adapt to larger disturbances while ensuring accurate velocity tracking.

In the early stages, researchers employed a simplified model known as the WIP [16] to approximate the dynamic characteristics of two-wheeled self-balance robots. Robust control methods were then conducted to regulate wheel torques and enhance the robustness of robots against uncertain disturbances. Techniques such as H-infinity control [17], sliding mode control [18], adaptive control [19], and disturbance observers [20,21] were applied to the WIP robot, confirming their effectiveness in ensuring system stability within a specific range of uncertainties. However, these methods were confined to evaluating the current state of the system, thereby overlooking its future behavior. Considering non-minimum phase systems such as WIP, these methodologies were found inadequate in achieving optimal control efficacy.

With the advancement of computational technology, the utilization of model predictive control (MPC) has progressively gained extensive traction. MPC operates by predicting the dynamic behavior of a system within a finite horizon and solves optimization control problems online to obtain optimal control input. Thus, MPC improves the transient response of the system, contributing to its widespread application in wheeled biped robots. Zafar et al. [22] utilized the nonlinear coupling between the pendulum and wheels in WIP, and employed differential dynamic programming to plan and map the CoM trajectory into joint space using whole-body control (WBC), achieving accurate velocity tracking in simulations. Klemm et al. [13] linearized the coupling relationship and used infinite-horizon linear quadratic regulator (LQR) to plan the CoM trajectory. However, these methods rely on accurate dynamic models. Thus, in the presence of modeling errors and uncertain disturbances, the velocity tracking performance of nominal MPC will degrade, even failing to ensure system stability. Aiming to address such challenges, robust MPC has been used to ensure convergence by tightening state constraints [23,24]; however, its applicability in robots is limited by its computational demands. The integration of the stability criterion based on the control Lyapunov function into the inequality constraints of the MPC framework has been demonstrated to enhance system stability against disturbance, revealing its practical applications in real robotic systems [10,25]. Furthermore, adaptive MPC uses measurement data during control, combining system identification [26] or learning techniques [27] to facilitate online adjustment of the feasible parameter set of uncertain terms, gradually converging them toward their true values and obtaining a highly accurate model [28,29]. Thus, adaptive MPC has been proven to enhance system robustness against uncertain terms over a broad range. However, this model has not yet been applied to wheeled biped robots; therefore, its potential to enhance the disturbance rejection performance of these robots requires further exploration.

Based on related works, an adaptive MPC framework for wheeled biped robots is developed by leveraging their under-actuated and coupling characteristics. This framework enables the robot to estimate and adapt to uncertain disturbances within a sagittal plane while maintaining accurate velocity tracking. The main contributions of this paper can be summarized as follows:

1) The linear coupling characteristics of wheeled biped robots under uncertain disturbances are revealed using a novel wheeled linear inverted pendulum (WLIP) model, achieving a unified expression of the comprehensive effects of disturbances on the system (Section 2.2).

2) On the basis of the CoM–wheel coupling relationship, a Kalman filter is used to estimate the CoM of the robot and floating base states as well as disturbances, integrating the information from joint encoders and inertial measurement unit (IMU) without relying on any external F/T sensors (Section 3.1).

3) A disturbance adaptive MPC framework for

wheeled biped robots is introduced, which allows the robot to adapt to various forms of external disturbances in the sagittal plane, ensuring stable motion and accurate velocity tracking (Section 3.2).

The proposed framework has been validated on a hydraulically driven wheeled biped robot, WLR-IV, and the experimental results are presented and discussed in Section 4. The paper is concluded with potential future works in Section 5.

2 Robot dynamic description and simplified modeling

The control method proposed in this paper employs a hierarchical architecture, including a CoM planner based on simplified model and a whole-body controller based on full-order model. In this section, the full-order dynamic model and the simplified WLIP model of the robot are individually established. The inherent dynamic coupling of the wheeled biped robot is then revealed through an analysis of the dynamic characteristics of the simplified model.

2.1 Robot dynamic

The objective of this study involves a class of wheeled biped robots, whose structure and degree of freedom (DoF) configuration are depicted in Fig. 1(a). These robotic systems are distinguished by their characteristic feature, where robot contact with the ground is maintained using a pair of wheels, revealing constrained axes of wheels within the same vertical plane during typical locomotion.

As shown in Fig. 1(a), an inertial coordinate frame $\{\mathcal{I}\}$ is attached to the ground, and a base coordinate frame $\{\mathcal{B}\}$ is attached to the pelvis. The origin of robot coordinate frame $\{\mathcal{R}\}$ coincides with $\{\mathcal{B}\}$, with the z-axis point vertically upward, and the x-axis is aligned with the forward direction of the robot. The generalized coordinates characterizing the robot system are defined as follows:

$$\boldsymbol{q} = \left[\boldsymbol{p}_{\mathrm{b}}^{\top}, \boldsymbol{q}_{\mathrm{b}}^{\top}, \boldsymbol{q}_{\mathrm{a}}^{\top}\right]^{\top} \in \mathrm{SE}(3) \times \mathbb{R}^{n_{\mathrm{a}}}, \tag{1}$$

where $\left[\boldsymbol{p}_{\mathrm{b}}^{\top}, \boldsymbol{q}_{\mathrm{b}}^{\top}\right]^{\top} \in \mathrm{SE}(3)$ represents the floating DoFs of the robot, and $\boldsymbol{q}_{\mathrm{a}} \in \mathbb{R}^{n_{\mathrm{a}}}$ represents the actuation DoFs of the robot with a count of n_{a}. $\boldsymbol{\tau} \in \mathbb{R}^{n_{\mathrm{a}}}$ represents the actuated vector. In the context of a wheeled biped robot, the knee, hip pitch, and wheel DoFs are considered indispensable. The dynamic equation of the robot can be expressed in the following form:

$$\begin{aligned} &\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{G}(\boldsymbol{q}) = \boldsymbol{S}^{\top}\boldsymbol{\tau} + \boldsymbol{J}_{\mathrm{c}}^{\top}\boldsymbol{P}\boldsymbol{\beta} + \boldsymbol{J}_{\mathrm{ext}}^{\top}\boldsymbol{F}_{\mathrm{ext}}, \\ &\boldsymbol{P} = \begin{bmatrix} \rho_{l1} \cdots \rho_{l4} & \boldsymbol{0} \\ \boldsymbol{0} & \rho_{r1} \cdots \rho_{r4} \end{bmatrix}, \end{aligned} \tag{2}$$

where $\boldsymbol{M}(\boldsymbol{q}) \in \mathbb{R}^{(n_{\mathrm{a}}+6)\times(n_{\mathrm{a}}+6)}$ is the positive definite symmetry inertia matrix, $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n_{\mathrm{a}}+6}$ is related to

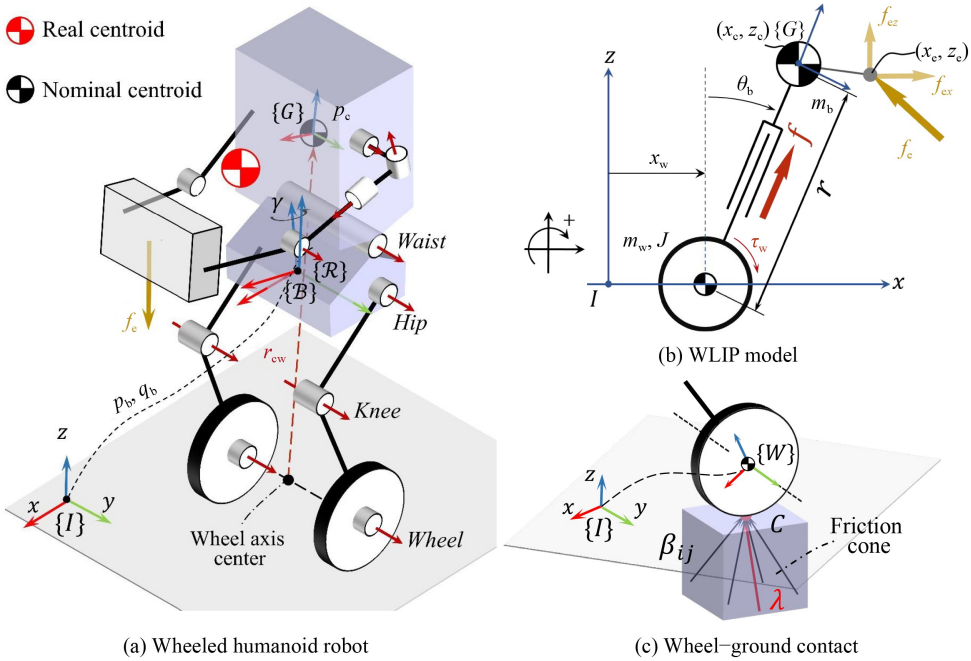


Fig. 1 (a) Typical configurations of wheeled biped robots. The wheel axis center is defined as the midpoint of the line connecting the centers of the two wheels. (b) Simplified WLIP model. (c) Wheel–ground contact model. The GRF $\boldsymbol{\lambda}$ is represented using four basis vectors $\boldsymbol{\rho}_{ij}$ $(i = l, r; j = 1, \ldots, 4)$ that approximate the friction cone, while β_{ij} is the magnitude in the direction of $\boldsymbol{\rho}_{ij}$.

the Coriolis and centrifugal forces, $\boldsymbol{G}(\boldsymbol{q}) \in \mathbb{R}^{n_a+6}$ is the gravity vector, and $\boldsymbol{S} \in \mathbb{R}^{n_a \times (n_a+6)}$ is the input selection matrix. $\boldsymbol{\beta} = [\beta_{l1}, ..., \beta_{l4}, \beta_{r1}, ..., \beta_{r4}]^\top$ gathers the magnitude of the friction cone basis vectors, and $\boldsymbol{J}_c \in \mathbb{R}^{6 \times (n_a+6)}$ is the contact Jacobian. Furthermore, $\boldsymbol{F}_{ext} \in \mathbb{R}^{n_f}$ is the external force vector, and $\boldsymbol{J}_{ext} \in \mathbb{R}^{n_f \times (n_a+6)}$ is its Jacobian.

Notably, the contact Jacobian in this study does not feature a full row rank, with a rank of 5, and a nonzero constant c exists such that ${}^{\mathcal{R}}\boldsymbol{J}_{c(1)} = c \cdot {}^{\mathcal{R}}\boldsymbol{J}_{c(5)}$, where ${}^{\mathcal{R}}\boldsymbol{J}_c$ is the contact Jacobian expressed in frame $\{\mathcal{R}\}$, and the subscripts are its corresponding rows. Therefore, the contact force cannot fully constrain the robot's spatial DoF, leading to a linear dependency between its forward motion and pitch rotation. This dependency lies at the core of the intrinsic instability and strong coupling characteristics exhibited by this robot type. Subsequently, a simplified model will be used to reveal this coupling and assess the effect of disturbances in the sagittal plane on the system.

2.2 Wheeled linear inverted pendulum

In contrast to conventional WIP robots, the wheeled biped robots represent a multi-DoF system that can modulate the distance between its CoM and wheels across a broad range. Aiming to characterize this feature, a prismatic joint is introduced to connect the CoM and wheel, with an actuated force f, thereby providing a simplified description of the robot's dynamic behavior using a variable-length WIP model, as depicted in Fig. 1(b). In this figure, (x_c, z_c) is the position of the nominal CoM, and the related position between the CoM and wheel is presented as $[x_{cw}, z_{cw}]^\top$.

The modeling of the variable-length WIP is constrained into the sagittal plane, wherein the CoM coordinate system $\{\mathcal{G}\}$ of the robot is attached to the nominal CoM, with the z-direction parallel to the prismatic joints. We denote $\boldsymbol{q}_w = [x_w, r, \theta_b]^\top$ as the generalized coordinates of the model, where x_w is the wheel position, r is the pendulum length, and θ_b is its angle of inclination. $\boldsymbol{F} = [\tau_w, f]^\top$ is the control input vector, where τ_w is the wheel torque. The disturbances affecting the robot are uniformly described in terms of a force $\boldsymbol{f}_e$ that acts at the point $\boldsymbol{p}_e = [x_e, z_e]^\top$, described in frame $\{\mathcal{G}\}$. Assuming pure rolling of the wheel on a flat ground surface, the dynamic equations of WLIP can be expressed as follows:

$$\boldsymbol{M}_w(\boldsymbol{q}_w)\ddot{\boldsymbol{q}}_w + \boldsymbol{C}_w(\boldsymbol{q}_w, \dot{\boldsymbol{q}}_w) = \boldsymbol{S}_w^\top \boldsymbol{F} + \boldsymbol{J}_e^\top \boldsymbol{f}_e, \tag{3}$$

where $\boldsymbol{M}_w(\boldsymbol{q}_w) \in \mathbb{R}^{3\times3}$, $\boldsymbol{C}_w(\boldsymbol{q}_w, \dot{\boldsymbol{q}}_w) \in \mathbb{R}^{3\times3}$, $\boldsymbol{S}_w \in \mathbb{R}^{2\times3}$, and $\boldsymbol{J}_e \in \mathbb{R}^{2\times3}$ are given by the following expressions:

$$\boldsymbol{M}_w = \begin{bmatrix} m_w + m_b + J/R^2 & m_b \sin\theta_b & m_b r\cos\theta_b \\ m_b \sin\theta_b & m_b & 0 \\ m_b r\cos\theta_b & 0 & m_b r^2 \end{bmatrix},$$

$$\boldsymbol{C}_w = \begin{bmatrix} -m_b r\dot{\theta}_b^2 \sin\theta_b + 2m_b \dot{r}\dot{\theta}_b \cos\theta_b \\ -m_b r\dot{\theta}_b^2 + m_b g\cos\theta_b \\ 2m_b r\dot{r}\dot{\theta}_b - m_b g r\sin\theta_b \end{bmatrix},$$

$$\boldsymbol{S}_w = \begin{bmatrix} -R^{-1} & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}^\top,$$

$$\boldsymbol{J}_e = \begin{bmatrix} 1 & \sin\theta_b & r\cos\theta_b - x_e\sin\theta_b + z_e\cos\theta_b \\ 0 & \cos\theta_b & -r\sin\theta_b - x_e\cos\theta_b - z_e\sin\theta_b \end{bmatrix},$$

$$\boldsymbol{f}_e = \begin{bmatrix} f_{ex} \\ f_{ez} \end{bmatrix},$$

where m_w, J, and R denote the mass, inertia, and radius of the wheel, respectively. m_b denotes the mass of the upper body.

With the introduction of variable length, the dynamic equations of the simplified model exhibit highly nonlinear, potentially impeding computational efficiency of motion planning. Aiming to improve the linearity of the model, considering the linear inverted pendulum model [30], the motion of the robot's CoM is confined to the horizontal x-y plane (i.e., $z_{cw} = r\cos\theta_b$ remain constant). Under this constraint, the horizontal position of the CoM and its second derivative are given by the following:

$$x_c = x_w + r\sin\theta_b, \tag{4a}$$

$$\ddot{x}_c = \ddot{x}_w + (\ddot{r} - r\dot{\theta}_b^2)\sin\theta_b + 2\dot{r}\dot{\theta}_b + r\ddot{\theta}_b. \tag{4b}$$

Based on the second line of the dynamic Eq. (3), the second term on the right-hand side of Eq. (4b) satisfies the following relationship:

$$\ddot{r} - r\dot{\theta}_b^2 = \frac{f}{m_b} - \ddot{x}_1\sin\theta_b - g\cos\theta_b + \frac{f_{ex}}{m_b}\sin\theta_b + \frac{f_{ez}}{m_b}\cos\theta_b. \tag{5}$$

Similarly, according to the third line of the dynamic Eq. (3), the third term on the right-hand side of Eq. (4b) satisfies:

$$\begin{aligned} 2\dot{r}\dot{\theta}_b + r\ddot{\theta}_b = &\frac{\tau}{m_b r} - \ddot{x}_1\cos\theta_b + g\sin\theta_b + \frac{f_{ex}}{m_b}\cos\theta_b \\ &- \frac{f_{ez}}{m_b}\sin\theta_b + \frac{\boldsymbol{f}_e^\top \boldsymbol{\Lambda}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{G}}\boldsymbol{p}_e}{m_b r}, \end{aligned} \tag{6}$$

where $\boldsymbol{\Lambda} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. ${}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{G}} = \begin{bmatrix} \cos\theta_b & -\sin\theta_b \\ \sin\theta_b & \cos\theta_b \end{bmatrix}$ is the rotation matrix from $\{\mathcal{I}\}$ to $\{\mathcal{G}\}$. Thus, through the substitution of Eqs. (5) and (6) into Eq. (4b), the relationship between the CoM horizontal acceleration and the F/T is given by the following:

$$\ddot{x}_c = \frac{f\sin\theta_b}{m_b} + \frac{\tau_w\cos\theta_b}{m_b r} + \frac{f_{ex}}{m_b} + \frac{\boldsymbol{f}_e^\top \boldsymbol{\Lambda}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{G}}\boldsymbol{p}_e}{m_b r}\cos\theta_b. \tag{7}$$

Furthermore, aiming to control CoM movement at a constant height, the prismatic joint force must balance the gravity, disturbance forces, and wheel torque of the robot, satisfying the following relationship:

$$f\cos\theta_b = \frac{\tau_\mathrm{w}}{r} + \frac{\sin\theta_b}{r}\boldsymbol{f}_\mathrm{e}^\top \boldsymbol{\Lambda}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{G}}\boldsymbol{p}_\mathrm{e} + m_\mathrm{b}g - f_{ez}. \tag{8}$$

Substituting Eq. (4b) into the first line of the dynamic equation, the wheel torque satisfies the following relationship:

$$\tau_\mathrm{w} = -(m_\mathrm{b}R + J/R)\ddot{x}_\mathrm{w} - m_\mathrm{b}R\ddot{x}_\mathrm{c} + f_{ex}R. \tag{9}$$

Considering $\cos\theta_\mathrm{b} = z_\mathrm{cw}/r$ and $\sin\theta_\mathrm{b} = x_\mathrm{cw}/z_\mathrm{cw}$, the combination of Eqs. (7)–(9) helps eliminate the control input f and τ_w, resulting in the dynamic relationship of the CoM and wheel under the effect of external forces:

$$\underbrace{(\tilde{M} + m_\mathrm{b}Rz_\mathrm{cw})}_{\eta}\ddot{x}_\mathrm{w} = -\underbrace{(m_\mathrm{b}R^2 + m_\mathrm{b}Rz_\mathrm{cw})}_{\alpha}\ddot{x}_\mathrm{cw} + \underbrace{m_\mathrm{b}gR}_{\mu}x_\mathrm{cw} + \underbrace{f_{ex}R^2 + \tau_\mathrm{e}R}_{\delta_\mathrm{d}}, \tag{10}$$

where τ_e is the equivalent torque of the disturbance forces acting on the wheel center:

$$\tau_\mathrm{e} = f_{ex}r\cos\theta_\mathrm{b} - f_{ez}r\sin\theta_\mathrm{b} + \boldsymbol{f}_\mathrm{e}^\top \boldsymbol{\Lambda}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{G}}\boldsymbol{p}_\mathrm{e}. \tag{11}$$

Similarly, the disturbance caused by an uncertain sloped terrain on the system can also be described as the torque residual at the wheel center between the actual and nominal GRF. Thus, the unified expression for the comprehensive effect of uncertain disturbances on the system can be obtained as follows:

$$\delta_\mathrm{d} = \begin{cases} f_{ex}R^2 + \tau_\mathrm{e}R, & \text{uncertain payload,} \\ (\lambda_z\sin\xi + \lambda_x(\cos\xi - 1))R^2, & \text{uneven terrain,} \end{cases} \tag{12}$$

where ξ is the ground slope. λ_x and λ_z are the GRF components in the forward and vertical directions, respectively.

As shown in Eq. (12), by imposing constraints on the CoM movement within a constant height, the dynamic relationship between the CoM and wheel is reduced into a linear coupling. This fundamental dynamic coupling is crucial to the control framework proposed in this paper. The following section will use this coupling relationship to estimate uncertain disturbances and regulate forward velocity through linear estimation and control techniques.

3 Disturbance adaptive model predictive controller

In this section, the disturbance adaptive MPC framework built upon the CoM–wheel linear coupling relationship is proposed, aiming to achieve robot states and disturbance estimation in the absence of external F/T sensors, thus enabling the robot to maintain accurate velocity tracking under uncertain disturbances. The architecture of the framework is shown in Fig. 2, which comprises a state and disturbance estimator based on Kalman filtering, a CoM planner based on MPC for forward velocity regulation, and a whole-body controller based on inverse dynamics control, generating the whole-body motion of the robot.

3.1 Disturbance estimation

The state of the robot's floating base includes position $\boldsymbol{p}_\mathrm{b}$, orientation $\boldsymbol{R}_\mathrm{b}$, linear velocity $\boldsymbol{v}_\mathrm{b}$, and angular velocity $\boldsymbol{\omega}_\mathrm{b}$. Among these, $\boldsymbol{\omega}_\mathrm{b}$ can be directly measured using a gyroscope, and the orientation estimation technology is relatively mature; thus, it has been integrated into most commercially available

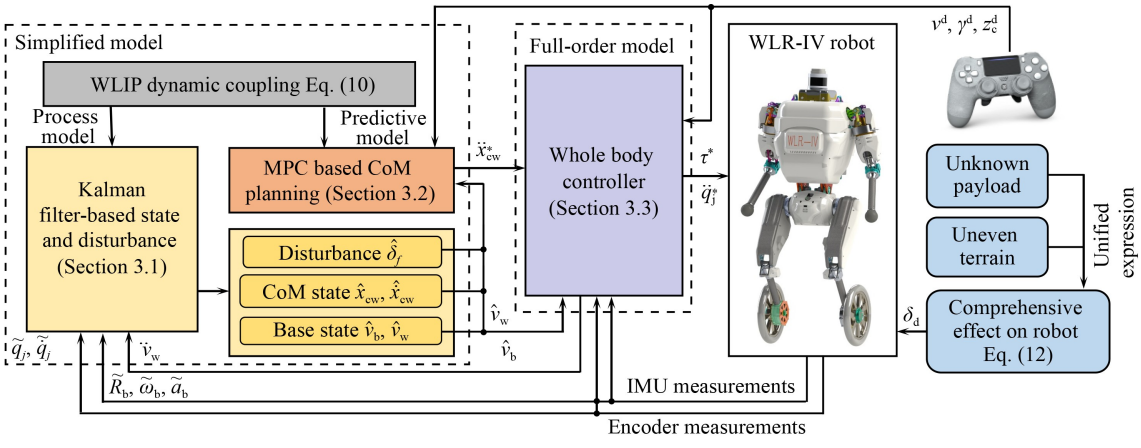


Fig. 2 Architecture of the disturbance adaptive model predictive controller for wheeled biped robots.

IMUs, increasing its accessibility. This work focuses on accurate velocity control of the robot. However, accurate position estimation typically relies on measurements from external sensing modules such as radars and cameras, which are beyond the scope of this work. Therefore, this study only involves linear velocity estimation of the robot's floating base. Accordingly, the state variables for the Kalman filter are determined as follows:

$$\boldsymbol{x}_{\mathrm{e}} = \mathrm{col}\big(\boldsymbol{v}_{\mathrm{b}}, \boldsymbol{v}_{\mathrm{w}}, x_{\mathrm{cw}}, \dot{x}_{\mathrm{cw}}, \delta_{\mathrm{d}}\big) \in \mathbb{R}^9,$$

where $\boldsymbol{v}_{\mathrm{w}} \in \mathbb{R}^3$ denotes the linear velocity of the wheel axis center, expressed in frame $\{\mathcal{I}\}$. The input is defined as follows:

$$\boldsymbol{u}_{\mathrm{e}} = \mathrm{col}\big(\boldsymbol{a}_{\mathrm{w}}, \boldsymbol{R}_{\mathrm{b}}\boldsymbol{\alpha}_{\mathrm{b}} + \boldsymbol{g}\big) \in \mathbb{R}^6,$$

where $\boldsymbol{a}_{\mathrm{w}} \in \mathbb{R}^3$ is the acceleration of the wheel axis center, which can be obtained using the optimal solution of WBC (Section 3.3). $\boldsymbol{\alpha}_{\mathrm{b}} \in \mathbb{R}^3$ is the acceleration of the floating base, directly measured using the accelerometer. $\boldsymbol{g} = \big[0, 0, -9.81\big]^\top \ \mathrm{m/s}^2$ is the gravity acceleration. The disturbances affecting the robot are random and unpredictable; thus, these disturbances are considered constant in the prediction process. Using the kinematic evolution and dynamic relationship from Eq. (10), a set of continuous time differential equations can be formulated as follows:

$$\begin{aligned}
&\frac{\mathrm{d}\boldsymbol{v}_{\mathrm{b}}}{\mathrm{d}t} = \boldsymbol{R}_{\mathrm{b}}\boldsymbol{\alpha}_{\mathrm{b}} + \boldsymbol{g},\\
&\frac{\mathrm{d}\boldsymbol{v}_{\mathrm{w}}}{\mathrm{d}t} = \boldsymbol{a}_{\mathrm{w}},\\
&\frac{\mathrm{d}x_{\mathrm{cw}}}{\mathrm{d}t} = \dot{x}_{\mathrm{cw}},\\
&\frac{\mathrm{d}\dot{x}_{\mathrm{cw}}}{\mathrm{d}t} = -\frac{\eta}{\alpha}\big[1,0,0\big]^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}}^\top\dot{\boldsymbol{v}}_{\mathrm{w}} + \frac{\mu}{\alpha}x_{\mathrm{cw}} + \frac{1}{\alpha}\delta_{\mathrm{d}},\\
&\frac{\mathrm{d}\delta_{\mathrm{d}}}{\mathrm{d}t} = 0,
\end{aligned} \tag{13}$$

where ${}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}}$ represents the rotation matrix from the inertia frame $\{\mathcal{I}\}$ to the robot frame $\{\mathcal{R}\}$, which can be obtained from the measurement of the base's yaw Euler angle γ:

$${}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}} = \boldsymbol{R}_z(\gamma).$$

The selection of measurements depends on the available sensor information for the robot. The state of the robot's actuation joints $\tilde{\boldsymbol{q}}_{\mathrm{a}}$ and $\tilde{\dot{\boldsymbol{q}}}_{\mathrm{a}}$ can be obtained from information of encoders and their derivatives, where the superscript $\tilde{(\cdot)}$ represents the measured value. Assuming that during the robot's motion, the wheels and the ground satisfy the pure rolling constraint, the velocity at the contact point is $\boldsymbol{0}$ in frame $\{\mathcal{I}\}$, and satisfied using the following constraint:

$$\boldsymbol{J}_{\mathrm{c}}\big[\boldsymbol{v}_{\mathrm{b}}^\top, \boldsymbol{\omega}_{\mathrm{b}}^\top, \boldsymbol{q}_{\mathrm{a}}^\top\big]^\top = \boldsymbol{0}. \tag{14}$$

Therefore, the linear velocity of the floating base according to left and right legs can be calculated from the measurable generalized velocities:

$$\begin{bmatrix}\tilde{\boldsymbol{v}}_{\mathrm{b}}^{\mathrm{l}}\\ \tilde{\boldsymbol{v}}_{\mathrm{b}}^{\mathrm{r}}\end{bmatrix} = -\boldsymbol{J}_{\mathrm{c}}^{0:2-1}\boldsymbol{J}_{\mathrm{c}}^{3:n_{\mathrm{a}}+5}\big[\tilde{\boldsymbol{\omega}}_{\mathrm{b}}^\top, \tilde{\dot{\boldsymbol{q}}}_{\mathrm{a}}^\top\big]^\top, \tag{15}$$

where the superscript $\{\cdot\}^{m:n}$ represents the matrix block formed by the mth to the nth columns.

The relative velocity between the wheel axis center and the base frame $\tilde{\dot{\boldsymbol{r}}}_{\mathrm{bw}}$, as well as the relative position $\tilde{\boldsymbol{r}}_{\mathrm{cw}}$ and velocity $\tilde{\dot{\boldsymbol{r}}}_{\mathrm{cw}}$ between the CoM and the wheel axis center, can also be measured by combining the measurable generalized states with kinematic model:

$$\begin{aligned}
&\tilde{\dot{\boldsymbol{r}}}_{\mathrm{bw}} = J_{\mathrm{bw}}^{3:n_{\mathrm{a}}+5}\big[\tilde{\boldsymbol{\omega}}_{\mathrm{b}}^\top, \tilde{\dot{\boldsymbol{q}}}_{\mathrm{a}}^\top\big]^\top,\\
&\dot{x}_{\mathrm{cw}} = \big[1,0,0\big]^{\mathcal{I}}\tilde{\boldsymbol{R}}_{\mathcal{R}}\boldsymbol{r}_{\mathrm{cw}}(\tilde{\boldsymbol{q}}_{\mathrm{a}}),\\
&\tilde{\dot{x}}_{\mathrm{cw}} = \big[1,0,0\big]^{\mathcal{I}}\tilde{\boldsymbol{R}}_{\mathcal{R}}\boldsymbol{J}_{\mathrm{cw}}^{3:n_{\mathrm{a}}+5}\big[\tilde{\boldsymbol{\omega}}_{\mathrm{b}}^\top, \tilde{\dot{\boldsymbol{q}}}_{\mathrm{a}}^\top\big]^\top.
\end{aligned} \tag{16}$$

Assuming that the mass of the robot is concentrated in the torso and pelvis, during stable motion, the CoM and the IMU lie on the same rigid body and maintain a constant relative position in the base coordinate frame. Consequently, the CoM linear acceleration measurement $\tilde{\ddot{x}}_{\mathrm{c}}$ can be approximated by incorporating the data $\tilde{\boldsymbol{a}}_{\mathrm{b}}$ from the accelerometer located on the floating base and the CoM-based kinematical model $\boldsymbol{r}_{\mathrm{bc}}(\tilde{\boldsymbol{q}})$:

$$\tilde{\ddot{x}}_{\mathrm{c}} \approx \big[1,0,0\big]\big(\tilde{\boldsymbol{R}}_{\mathrm{b}}\big(\tilde{\boldsymbol{a}}_{\mathrm{b}} + \tilde{\dot{\boldsymbol{\omega}}}_{\mathrm{b}} \times \boldsymbol{r}_{\mathrm{bc}} + \tilde{\boldsymbol{\omega}}_{\mathrm{b}} \times (\tilde{\boldsymbol{\omega}}_{\mathrm{b}} \times \boldsymbol{r}_{\mathrm{bc}})\big)\big). \tag{17}$$

This measurement is an approximation; thus, a large measurement noise covariance is assigned to reflect a low confidence level. The process noise covariance matrix $\boldsymbol{Q}_{\mathrm{e}}$ and the measurement noise covariance matrix $\boldsymbol{R}_{\mathrm{e}}$ in the Kalman filter are both diagonal matrices, and their elements are listed in Tables S1 and S2 of the Electronic Supplementary Material.

In the prediction step of the Kalman filter, the *a posteriori* estimated state $\hat{\boldsymbol{x}}_{\mathrm{e},k}^{+}$ at time step k is propagated to the next time step on the differentiable manifold from Eq. (13), yielding the prior state estimate $\hat{\boldsymbol{x}}_{\mathrm{e},k+1}^{-}$ at time step:

$$\hat{\boldsymbol{x}}_{\mathrm{e},k+1}^{-} = \boldsymbol{A}_{\mathrm{e}}\hat{\boldsymbol{x}}_{\mathrm{e},k}^{+} + \boldsymbol{B}_{\mathrm{e},k}\boldsymbol{u}_{\mathrm{e},k},$$

$$\boldsymbol{A}_{\mathrm{e}} = \begin{bmatrix}
\mathbb{I}_3 & \boldsymbol{0}_{3\times3} & 0 & 0 & 0\\
\boldsymbol{0}_{3\times3} & \mathbb{I}_3 & 0 & 0 & 0\\
\boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & 1+\dfrac{\mu}{2\alpha}\Delta t^2 & \Delta t & \dfrac{1}{2\alpha}\Delta t^2\\
\boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & \dfrac{\mu}{\alpha}\Delta t & 1 & \dfrac{1}{\alpha}\Delta t\\
\boldsymbol{0}_{1\times3} & \boldsymbol{0}_{1\times3} & 0 & 0 & 1
\end{bmatrix},$$

$$\boldsymbol{B}_{\mathrm{e},k}=\begin{bmatrix}\boldsymbol{0}_{3\times3} & \Delta t\mathbb{I}_3\\ \Delta t\mathbb{I}_3 & \boldsymbol{0}_{3\times3}\\ -\dfrac{\eta}{2\alpha}[1,0,0]^{\mathcal{I}}\tilde{\boldsymbol{R}}_{\mathcal{R}}\Delta t^2 & \boldsymbol{0}_{3\times3}\\ -\dfrac{\eta}{\alpha}[1,0,0]^{\mathcal{I}}\tilde{\boldsymbol{R}}_{\mathcal{R}}\Delta t & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\end{bmatrix}, \tag{18}$$

where Δt is the sampling period of the controller, and $\mathbb{I}_n$ denotes an identity matrix of order n. An *a priori* estimate of the covariance matrix at the time step $k+1$ can be obtained from the posteriori estimate of the state covariance matrix $\boldsymbol{P}_k$ at the previous time step:

$$\boldsymbol{P}_{k+1}^{-}=\boldsymbol{A}_{\mathrm{e}}\boldsymbol{P}_k\boldsymbol{A}_{\mathrm{e}}^{\top}+\boldsymbol{Q}_{\mathrm{e},k}. \tag{19}$$

Considering the observed inaccuracies in the prediction process, using measurements to correct the prior state estimate is necessary to obtain the optimal estimated states. Based on previous discussion, the measurement vector is chosen as follows:

$$\boldsymbol{y}_{\mathrm{e}}=\mathrm{col}\big(\boldsymbol{v}_{\mathrm{b}}^{\mathrm{l}},\boldsymbol{v}_{\mathrm{b}}^{\mathrm{r}},\dot{\boldsymbol{r}}_{\mathrm{bw}},\ddot{x}_{\mathrm{c}},x_{\mathrm{cw}},\dot{x}_{\mathrm{cw}}\big)\in\mathbb{R}^{12}.$$

The mapping relationship between the measurement vector and the estimated states is as follows:

$$\begin{aligned}
&\boldsymbol{y}_{\mathrm{e},k}=\boldsymbol{\Psi}_{\mathrm{e}}\boldsymbol{x}_{\mathrm{e},k}+\boldsymbol{L}_{\mathrm{e},k}\boldsymbol{u}_{\mathrm{e},k},\\
&\boldsymbol{\Psi}_{\mathrm{e}}=\begin{bmatrix}\mathbb{I}_{3\times3} & \boldsymbol{0}_{3\times3} & 0 & 0 & 0\\ \mathbb{I}_{3\times3} & \boldsymbol{0}_{3\times3} & 0 & 0 & 0\\ \mathbb{I}_{3\times3} & -\mathbb{I}_{3\times3} & 0 & 0 & 0\\ \boldsymbol{0}_{3\times1} & \boldsymbol{0}_{3\times1} & \dfrac{\beta}{\alpha} & 0 & \dfrac{1}{\alpha}\\ \boldsymbol{0}_{3\times1} & \boldsymbol{0}_{3\times1} & 1 & 0 & 0\\ \boldsymbol{0}_{3\times1} & \boldsymbol{0}_{3\times1} & 0 & 1 & 0\end{bmatrix},\\
&\boldsymbol{L}_{\mathrm{e},k}=\begin{bmatrix}\boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times3} & \boldsymbol{0}_{3\times3}\\ \boldsymbol{0}_{3\times1} & \dfrac{\alpha-\mu}{\alpha}[1,0,0]\tilde{\boldsymbol{R}}_{\mathrm{b}}\\ \boldsymbol{0}_{2\times3} & \boldsymbol{0}_{2\times3}\end{bmatrix}.
\end{aligned} \tag{20}$$

Finally, the *a posterior* state estimate $\hat{\boldsymbol{x}}_{\mathrm{e},k}^{+}$ at time step k can be obtained using the following correction process of the standard Kalman filter:

$$\begin{aligned}
&\boldsymbol{S}_k=\boldsymbol{\Psi}_{\mathrm{e}}\boldsymbol{P}_k^{-}\boldsymbol{\Psi}_e^{\top}+\boldsymbol{R}_{\mathrm{e},k},\\
&\boldsymbol{K}_k=\boldsymbol{P}_k^{-}\boldsymbol{\Psi}_{\mathrm{e}}^{\top}\boldsymbol{S}_k^{-1},\\
&\boldsymbol{P}_k=(\mathbb{I}_9-\boldsymbol{K}_k\boldsymbol{\Psi}_{\mathrm{e}})\boldsymbol{P}_k^{-},\\
&\hat{\boldsymbol{x}}_{\mathrm{e},k}^{+}=\hat{\boldsymbol{x}}_{\mathrm{e},k}^{-}+\boldsymbol{K}_k(\boldsymbol{y}_{\mathrm{e},k}-\boldsymbol{L}_{\mathrm{e},k}\boldsymbol{K}_k\boldsymbol{u}_{\mathrm{e},k}-\boldsymbol{\Psi}_{\mathrm{e}}\hat{\boldsymbol{x}}_{\mathrm{e},k}^{-}),
\end{aligned} \tag{21}$$

where $\boldsymbol{S}_k$ is the innovation covariance, and $\boldsymbol{K}_k$ is the Kalman gain. The estimated CoM states $\hat{x}_{\mathrm{cw}}$, $\hat{\dot{x}}_{\mathrm{cw}}$, wheel axis center velocity $\hat{\boldsymbol{v}}_{\mathrm{w}}$, and disturbances $\hat{\delta}_{\mathrm{d}}$ will serve as state feedback for the CoM planning MPC in the next section. Meanwhile, the base velocity $\hat{\boldsymbol{v}}_{\mathrm{b}}$ will be combined with other measurable state values to obtain the complete generalized state of the robot.

3.2 MPC-based CoM planning

The WLIP is characterized by its non-minimum phase behavior. In response to a rapid input signal, an inherent velocity undershoot is observed at the beginning of acceleration. This undershoot is necessary to reposition the CoM forward, facilitating the acceleration process. Aiming to effectively manage this characteristic, MPC is employed in this study to optimize the response of the system by considering its future behavior over a finite time horizon, and the forward velocity is regulated by planning the CoM motion of the robot based on its dynamic coupling. Furthermore, accounting for the effect of uncertain disturbances within the nominal MPC framework, this study aims to achieve robust motion and accurate velocity tracking for the robot under uncertain disturbances.

The state variables of MPC are chosen as $\boldsymbol{x}_{\mathrm{c}}=[x_{\mathrm{cw}},\dot{x}_{\mathrm{cw}},v]^{\top}\in\mathbb{R}^3$, where $v=\dot{x}_{\mathrm{w}}$ denotes the forward velocity of the robot. The control input $u_{\mathrm{c}}=\ddot{x}_{\mathrm{cw}}\in\mathbb{R}$ is set as the CoM–wheel-related horizontal acceleration. Based on the coupling relationship from Eq. (10), the discrete-time system model can be expressed as follows:

$$\begin{aligned}
&\boldsymbol{x}_{\mathrm{c},k+1}=\boldsymbol{G}\boldsymbol{x}_{\mathrm{c},k}+\boldsymbol{H}u_{\mathrm{c},k}+\boldsymbol{D}\hat{\delta}_{\mathrm{d},k},\\
&\boldsymbol{H}=\begin{bmatrix}1 & \delta t & 0\\ 0 & 1 & 0\\ \dfrac{\mu}{\eta}\delta t & 0 & 1\end{bmatrix},\\
&\boldsymbol{G}=\begin{bmatrix}0\\ \delta t\\ \dfrac{\alpha}{\eta}\delta t\end{bmatrix},\\
&\boldsymbol{D}=\begin{bmatrix}0\\ 0\\ \dfrac{1}{\eta}\delta t\end{bmatrix},
\end{aligned} \tag{22}$$

where δt is the predictive sampling period. Considering the dependence of the system model on the CoM height, it is updated at each control node and remains constant within the prediction horizon.

Using the disturbance estimator, the disturbances $\hat{\delta}_{\mathrm{d},k}$ are considered known and constant throughout the prediction sequence, and then updated at each control step. Leveraging the prediction model, the dynamic behavior in the future horizon can be predicted using the estimated states at time step k:

$$\boldsymbol{X}_k=\boldsymbol{F}\hat{x}_{\mathrm{c},k}+\boldsymbol{\Gamma}\boldsymbol{U}_k+\boldsymbol{\Pi}\hat{\delta}_{\mathrm{d},k}, \tag{23}$$

where $\boldsymbol{X}_k=\mathrm{col}\big(\boldsymbol{x}_{\mathrm{c},k+1|k},\ldots,\boldsymbol{x}_{\mathrm{c},k+N_{\mathrm{p}}|k}\big)$ represents the

predicted state sequence. The input sequence is represented as $\boldsymbol{U}_k = \mathrm{col}\big(u_{\mathrm{c},k|k}, \ldots, u_{\mathrm{c},k+N_\mathrm{p}-1|k}\big)$, and N_p is the prediction horizon. Furthermore, we have

$$\boldsymbol{F} = \begin{bmatrix} \boldsymbol{G} \\ \boldsymbol{G}^2 \\ \vdots \\ \boldsymbol{G}^{N_\mathrm{p}} \end{bmatrix},$$

$$\boldsymbol{\Gamma} = \begin{bmatrix} \boldsymbol{H} & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ \boldsymbol{G}\boldsymbol{H} & \boldsymbol{H} & \cdots & \boldsymbol{0} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{G}^{N_\mathrm{p}}\boldsymbol{H} & \boldsymbol{G}^{N_\mathrm{p}-1}\boldsymbol{H} & \cdots & \boldsymbol{H} \end{bmatrix},$$

$$\boldsymbol{\Pi} = \begin{bmatrix} \boldsymbol{D} & \boldsymbol{H}+\boldsymbol{D} & \cdots & \boldsymbol{H}^{N_\mathrm{p}-1}+\cdots+\boldsymbol{H}+\boldsymbol{D} \end{bmatrix}^\top.$$

The control objective of the MPC focuses on the convergence of the system to the equilibrium point and the tracking of the desired forward velocity v_k^d, while simultaneously minimizing energy consumption. This energy consumption can be traded off using a symmetric positive definite weight matrix, denoted as $\boldsymbol{Q}$ for tracking errors, and $\boldsymbol{R}$ for energy consumption. Thus, the cost can be defined as follows:

$$\begin{aligned} J_k &= \left\|\boldsymbol{X}_k - \boldsymbol{X}_k^\mathrm{d}\right\|_{\boldsymbol{Q}}^2 + \|\boldsymbol{U}_k\|_{\boldsymbol{R}}^2 J_k \\ &= \boldsymbol{U}_k^\top(\boldsymbol{\Gamma}^\top\boldsymbol{Q}\boldsymbol{\Gamma}+\boldsymbol{R})\boldsymbol{U}_k + 2(\boldsymbol{F}\hat{\boldsymbol{x}}_{\mathrm{c},k} + \boldsymbol{\Pi}\hat{\delta}_{\mathrm{d},k})^\top\boldsymbol{Q}\boldsymbol{\Gamma}\boldsymbol{U}_k, \end{aligned} \tag{24}$$

where the equation overlooks the constant term, and $\boldsymbol{X}_k^\mathrm{d} = \mathrm{col}(\boldsymbol{x}_{k+1|k}^\mathrm{d}, \ldots, \boldsymbol{x}_{k+N_\mathrm{p}|k}^\mathrm{d})$ represents the reference state trajectory, with each segment defined as $\boldsymbol{x}_{i|k}^\mathrm{d} = \left[0, 0, v_{i|k}^\mathrm{d}\right]^\top$, $i = k+1, \ldots, k+N_\mathrm{p}$.

Moreover, limited by the structural design and joint drive capabilities of the robot, restricting the CoM dynamic and forward velocity during movement is necessary. These constraints can be uniformly described as follows:

$$\begin{aligned} &\boldsymbol{X}_{\min} \leqslant \boldsymbol{X}_k \leqslant \boldsymbol{X}_{\max}, \\ &\boldsymbol{U}_{\min} \leqslant \boldsymbol{U}_k \leqslant \boldsymbol{U}_{\max}. \end{aligned}$$

Based on the above definition, the MPC-based motion velocity controller for wheeled biped robots can be formulated as the following standard dense quadratic programming (QP) problem:

$$\begin{aligned} \min_{\boldsymbol{U}(k)} \quad & \begin{bmatrix} \boldsymbol{U}_k \\ 1 \end{bmatrix}^\top \begin{bmatrix} \boldsymbol{\Gamma}^\top\boldsymbol{Q}\boldsymbol{\Gamma}+\boldsymbol{R} & \boldsymbol{F}\hat{\boldsymbol{x}}_{\mathrm{c},k} + \boldsymbol{\Pi}\hat{\delta}_{\mathrm{d},k} \\ \boldsymbol{F}\hat{\boldsymbol{x}}_{\mathrm{c},k} + \boldsymbol{\Pi}\hat{\delta}_{\mathrm{d},k} & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{U}_k \\ 1 \end{bmatrix}, \\ \text{s.t.} \quad & \boldsymbol{X}_{\min} \leqslant \boldsymbol{F}\hat{\boldsymbol{x}}_{\mathrm{c},k} + \boldsymbol{\Gamma}\boldsymbol{U}(k) + \boldsymbol{\Pi}\hat{\delta}_{\mathrm{d},k} \leqslant \boldsymbol{X}_{\max}, \\ & \boldsymbol{U}_{\min} \leqslant \boldsymbol{U}_k \leqslant \boldsymbol{U}_{\max}, \\ & \big[\boldsymbol{0}, \boldsymbol{0}, \ldots, \mathbb{I}_3\big](\boldsymbol{F}\hat{\boldsymbol{x}}_{\mathrm{c},k} + \boldsymbol{\Gamma}\boldsymbol{U}_k + \boldsymbol{\Pi}\hat{\delta}_{\mathrm{d},k}) \in \boldsymbol{\chi}_N, \end{aligned} \tag{25}$$

where $\boldsymbol{\chi}_N$ denotes the terminal invariant set, which ensures the feasibility and stability of the MPC controller by constraining the terminal states of the system. In this work, the terminal invariant set is tightened, and a terminal equality constraint is imposed to enhance the convergence speed of the system, thereby improving the velocity tracking performance of the robot; that is, the terminal constraint is set as $[\boldsymbol{0}, \boldsymbol{0}, \ldots, \mathbb{I}_3](\boldsymbol{F}\hat{\boldsymbol{x}}_{\mathrm{c},k} + \boldsymbol{\Gamma}\boldsymbol{U}_k + \boldsymbol{\Pi}\hat{\delta}_{\mathrm{d},k}) = \boldsymbol{x}_{k+N_\mathrm{p}|k}^\mathrm{d}$.

The QP problem can be easily solved using the open-source general-purpose QP solvers, and the first term of the optimized control input sequence $\ddot{x}_{\mathrm{cw},k}^* = u_{\mathrm{c},k+1|k}^*$ is subsequently implemented in the system. Notably, the control input, that is, the CoM–wheel related horizontal acceleration, cannot be directly applied to the robot. Instead, they must be tracked using a whole-body controller, which will be formulated in the forthcoming section.

3.3 Whole-body controller

In this section, the full-order multi-rigid body dynamics system of the robot will be considered for the optimal generation of robot joint inputs based on hierarchical inverse dynamic control. This system enables the robot to track the reference CoM dynamics generated by the MPC, while ensuring its movement stability.

Due to the limitations of the robot's DoFs, potential conflicts may arise during the operation of various tasks. As a solution, the tracking objectives of the task space can be formulated as QP problems that penalize the robot's deviation from desired behaviors $\|w_i(\boldsymbol{A}_{\mathrm{T},i}\boldsymbol{\zeta} - \boldsymbol{b}_{\mathrm{T},i})\|^2$ and prioritize them by applying weights w_i. In the formulation, $\boldsymbol{\zeta} = \mathrm{col}(\ddot{\boldsymbol{q}}, \boldsymbol{\tau}, \boldsymbol{\beta})$ are the optimization variables. Through the aggregation of multiple task priorities into a comprehensive task matrix, the joint motions can be efficiently generated by solving a single QP problem:

$$\begin{aligned} \min_{\boldsymbol{\zeta}} \quad & 0.5\boldsymbol{\zeta}^\top\boldsymbol{A}_\mathrm{T}^\top\boldsymbol{A}_\mathrm{T}\boldsymbol{\zeta} - \boldsymbol{b}_\mathrm{T}^\top\boldsymbol{A}_\mathrm{T}\boldsymbol{\zeta}, \\ \text{s.t.} \quad & \boldsymbol{C}_\mathrm{E}\boldsymbol{\zeta} + \boldsymbol{c}_\mathrm{E} = \boldsymbol{0}, \\ & \boldsymbol{C}_\mathrm{I}\boldsymbol{\zeta} + \boldsymbol{c}_\mathrm{I} \leqslant \boldsymbol{0}, \end{aligned} \tag{26}$$

where $\boldsymbol{A}_\mathrm{T} = \left[w_0\boldsymbol{A}_{\mathrm{T},0}^\top, \ldots, w_n\boldsymbol{A}_{\mathrm{T},n}^\top\right]^\top$ and $\boldsymbol{b}_\mathrm{T} = \left[w_0\boldsymbol{b}_{\mathrm{T},0}^\top, \ldots, w_n\boldsymbol{b}_{\mathrm{T},n}^\top\right]^\top$ are task matrices. $\boldsymbol{C}_\mathrm{E}$, $\boldsymbol{c}_\mathrm{E}$, $\boldsymbol{C}_\mathrm{I}$, and $\boldsymbol{c}_\mathrm{I}$ are the equality and inequality constraints that must be satisfied to maintain stable robot motion. In the following, the tasks will be described in order of high to low priority.

1) Dynamic constraint: Aiming to address the basic physical rules of robot motion, the joint acceleration, joint torques, and contact forces of the robot must satisfy the dynamics Eq. (2), that is, conform to the following equation constraint:

$$\begin{bmatrix} \boldsymbol{M}(\boldsymbol{q}) & -\boldsymbol{S}^\top & -\boldsymbol{J}_\mathrm{c}^\top(\boldsymbol{q})\boldsymbol{P} \end{bmatrix}\boldsymbol{\zeta} = -\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) - \boldsymbol{G}(\boldsymbol{q}). \tag{27}$$

2) Contact constraints: As shown in Fig. 1(c), the robot wheels are in point contact with the ground. Under the assumption of pure rolling, the velocity and acceleration of the contact point in the world coordinate system are zero. Thus, the contact constraint can be described using the following:

$$\begin{bmatrix}\boldsymbol{J}_{\mathrm{c}}(\boldsymbol{q}) & \mathbf{0} & \mathbf{0}\end{bmatrix}\boldsymbol{\zeta}+\dot{\boldsymbol{J}}_{\mathrm{c}}(\boldsymbol{q})\dot{\boldsymbol{q}}=\mathbf{0}. \tag{28}$$

Furthermore, aiming to prevent slippage between the robot and the ground, while still ensuring a stable wheel–ground contact, the contact force vector must be constrained within the friction cone:

$$\beta_{i,j}>0 \quad (i=l,r;\, j=1,\ldots,4). \tag{29}$$

3) Base posture task: The wheel–ground contact Jacobian of the wheeled biped is not full rank; thus, the contact wrench can only constrain five DoFs of the floating base, excluding the pitch angle. Hence, implementing additional control is necessary to govern the pitch angle of the base, thereby ensuring the stability of the robot's motion. Simultaneously, the roll angle control of the base becomes essential to enhance the lateral stability of the robot when traversing uneven terrains. Moreover, the base acceleration in the yaw direction must be equal to the steering acceleration generated by MPC, and the task is given as follows:

$$\begin{aligned}\boldsymbol{A}_{\mathrm{T},0}&=\begin{bmatrix}\boldsymbol{J}_{\mathrm{b}}^{\mathrm{r}} & \mathbf{0} & \mathbf{0}\end{bmatrix},\\ \boldsymbol{b}_{\mathrm{T},0}&=\boldsymbol{K}_{\mathrm{b}}\phi_{\mathrm{b}}^{\mathrm{err}}+\boldsymbol{D}_{\mathrm{b}}\boldsymbol{\omega}_{\mathrm{b}}^{\mathrm{err}}-\dot{\boldsymbol{J}}_{\mathrm{b}}^{\mathrm{r}}\dot{\boldsymbol{q}},\end{aligned} \tag{30}$$

where $\boldsymbol{J}_{\mathrm{b}}^{\mathrm{r}}\in\mathbb{R}^{3\times n_{\mathrm{a}}+6}$ denotes the rotation Jacobian of base, and $\phi_{\mathrm{b}}^{\mathrm{err}}$ denotes the angle error of the base, expressed in axis angle. The reference orientation of the base is set as $\boldsymbol{R}_{\mathrm{b}}^{\mathrm{d}}=\mathbb{I}_3$, while the reference angular velocity of the base is set as $\boldsymbol{\omega}_{\mathrm{b}}^{\mathrm{d}}=0$. $\boldsymbol{K}_{\mathrm{b}}$ and $\boldsymbol{D}_{\mathrm{b}}$ are positive definite gain matrices.

4) CoM tracking task: The CoM of the robot must track the reference acceleration trajectory generated by MPC, while simultaneously maintaining a constant height $z_{\mathrm{cw}}^{\mathrm{d}}$ to meet the constraints in WLIP. The corresponding task submatrix is given by the following:

$$\begin{aligned}\boldsymbol{A}_{\mathrm{T},1}&=\begin{bmatrix}\boldsymbol{J}_{\mathrm{cw}} & \mathbf{0} & \mathbf{0}\end{bmatrix},\\ \boldsymbol{b}_{\mathrm{T},1}&={}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}}(\ddot{\boldsymbol{r}}_{\mathrm{cw}}^{\mathrm{ref}}+\boldsymbol{K}_{\mathrm{c}}\boldsymbol{r}_{\mathrm{cw}}^{\mathrm{err}}+\boldsymbol{D}_{\mathrm{c}}\dot{\boldsymbol{r}}_{\mathrm{cw}}^{\mathrm{err}}-\dot{\boldsymbol{J}}_{\mathrm{cw}}\dot{\boldsymbol{q}}),\end{aligned} \tag{31}$$

where $\ddot{\boldsymbol{r}}_{\mathrm{cw}}^{\mathrm{ref}}=\begin{bmatrix}\ddot{x}_{\mathrm{cw}}^{*},0,0\end{bmatrix}^{\top}$ is the reference linear acceleration of the CoM relative to the wheel axis center, and $\ddot{x}_{\mathrm{cw}}^{*}$ is the optimal result of the CoM planning MPC. $\boldsymbol{r}_{\mathrm{cw}}^{\mathrm{err}}$ is the CoM–wheel relation position tracking error, where the desired CoM–wheel-related position is $\boldsymbol{r}_{\mathrm{cw}}^{\mathrm{d}}=\begin{bmatrix}0,0,z_{\mathrm{cw}}^{\mathrm{d}}\end{bmatrix}$. $z_{\mathrm{cw}}^{\mathrm{d}}$ is also set as the height of the CoM in WLIP, and the desired related velocity is set as $\mathbf{0}$. $\boldsymbol{J}_{\mathrm{cw}}\in\mathbb{R}^{3\times(n_{\mathrm{a}}+6)}$ is the relative velocity Jacobian of the CoM–wheel, expressed in frame $\{\mathcal{I}\}$. $\boldsymbol{K}_{\mathrm{c}}\in\mathbb{R}^{3\times3}$ and $\boldsymbol{D}_{\mathrm{c}}\in\mathbb{R}^{3\times3}$ are positive definite gain matrices whose elements in the upper-left corner are all zeros.

5) Wheel-relative position task: Aiming to ensure that the robot meets the dynamic behavior of the WLIP model, constraining the axes of the two wheels within a vertical plane is necessary. Consequently, controlling the relative positions of the two wheel frames in the direction of the robot's forward movement is also necessary, maintaining them around a value of zero. The corresponding task can be articulated as follows:

$$\begin{aligned}\boldsymbol{A}_{\mathrm{T},2}&=\begin{bmatrix}\begin{bmatrix}1 & 0 & 0\end{bmatrix}{}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}}^{\top}(\boldsymbol{J}_{\mathrm{lw}}^{\mathrm{l}}-\boldsymbol{J}_{\mathrm{rw}}^{\mathrm{l}}) & \mathbf{0} & \mathbf{0}\end{bmatrix},\\ \boldsymbol{b}_{\mathrm{T},2}&=\boldsymbol{K}_{\mathrm{w}}{}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}}^{\top}(\boldsymbol{p}_{\mathrm{lw}}-\boldsymbol{p}_{\mathrm{rw}})+\boldsymbol{D}_{\mathrm{w}}{}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}}^{\top}(\dot{\boldsymbol{p}}_{\mathrm{lw}}-\dot{\boldsymbol{p}}_{\mathrm{rw}})\\ &\quad-\begin{bmatrix}1 & 0 & 0\end{bmatrix}{}^{\mathcal{I}}\boldsymbol{R}_{\mathcal{R}}^{\top}(\dot{\boldsymbol{J}}_{\mathrm{lw}}^{\mathrm{l}}-\dot{\boldsymbol{J}}_{\mathrm{rw}}^{\mathrm{l}})\dot{\boldsymbol{q}},\end{aligned} \tag{32}$$

where $\boldsymbol{J}_{\mathrm{lw}}^{\mathrm{l}}$ and $\boldsymbol{J}_{\mathrm{rw}}^{\mathrm{l}}$ represent the linear Jacobians of the left and wheel frames, respectively. $\boldsymbol{p}_{\mathrm{lw}}$ and $\boldsymbol{p}_{\mathrm{rw}}$ are their positions in frame $\{\mathcal{I}\}$. $\boldsymbol{K}_{\mathrm{w}}=[k_{\mathrm{p,w}},0,0]$ and $\boldsymbol{D}_{\mathrm{w}}=[k_{\mathrm{d,w}},0,0]$ are feedback gain matrices.

6) Nominal trajectory penalty: Owing to the redundant DoFs, an infinite number of configurations will be used for an identical target. The undesirable body postures can be avoided by predefining a series of nominal joint configurations $\boldsymbol{q}_{\mathrm{nom}}$ and penalizing deviations in joint angles, simultaneously minimizing the joint torques:

$$\begin{aligned}\boldsymbol{A}_{\mathrm{T},2}&=\mathbb{I}_{2n_a+14},\\ \boldsymbol{b}_{\mathrm{T},2}&=\begin{bmatrix}\ddot{\boldsymbol{q}}_{\mathrm{nom}} & \mathbf{0} & \mathbf{0}\end{bmatrix}^{\top},\\ \ddot{\boldsymbol{q}}_{\mathrm{nom}}&=-\boldsymbol{K}_{\mathrm{q}}(\boldsymbol{q}-\boldsymbol{q}_{\mathrm{nom}})-\boldsymbol{D}_{\mathrm{q}}\dot{\boldsymbol{q}}.\end{aligned} \tag{33}$$

After solving the QP problem from Eq. (26), the optimal joint torques $\boldsymbol{\tau}^{*}$ will be sent to the underlying joint servos, and the joint acceleration $\ddot{\boldsymbol{q}}^{*}$ will be used to calculate $\boldsymbol{a}_{\mathrm{w}}$, which will then be used as the input of the state estimator:

$$\boldsymbol{a}_{\mathrm{w}}=\boldsymbol{J}_{\mathrm{w}}\ddot{\boldsymbol{q}}^{*}+\dot{\boldsymbol{J}}_{\mathrm{w}}\tilde{\dot{\boldsymbol{q}}}, \tag{34}$$

where $\boldsymbol{J}_{\mathrm{w}}$ is the linear velocity Jacobian of the wheel axis center, expressed in frame $\{\mathcal{I}\}$.

4 Results and discussion

Aiming to evaluate the performance of the control method proposed in this paper, experiments are conducted on a hydraulic wheeled biped robot WLR-IV, as shown in Fig. 3. WLR-IV has a height of 1.58 m and weighs approximately 65 kg. Each leg has three DoFs, including hip pitch, knee, and wheel DoF. Each arm includes a 3-DoF shoulder and a 1-DoF elbow. Additionally, the waist joint has 1 DoF in pitch. Excluding the wheels and shoulder yaw joints,

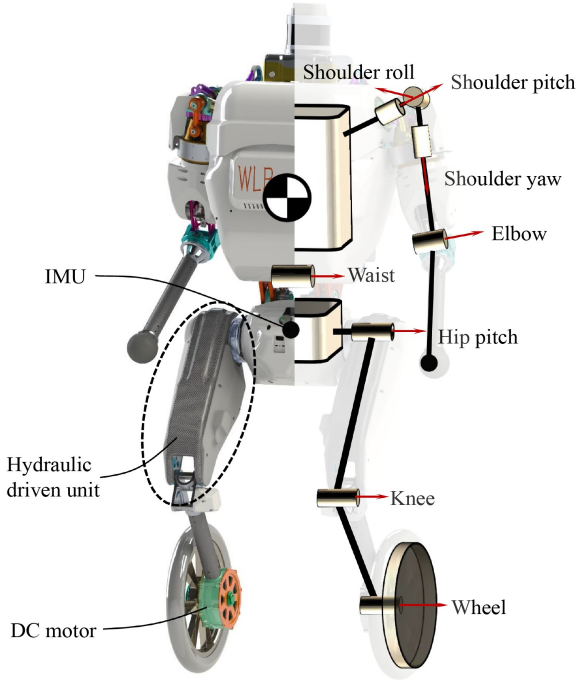


Fig. 3 Wheeled biped robot WLR-IV served as the experimental platform to evaluate the performance of the control method proposed in this paper.

which are driven by DC motors, all other joints are hydraulically driven. The IMU (Xsens MTI-G-710 model) is mounted on the pelvis and sends the orientation, angular velocity, and linear acceleration information through the serial port at a frequency of 400 Hz. Linear variable displacement transducers are used to measure the displacements of the linear hydraulic cylinders, which are then converted into joint angles using linkage kinematics. The rotation of the joint motors are directly measured using Renishaw RMB20 rotary magnetic encoder modules, and the joint angles are obtained by applying the reduction ratio. All computations were conducted using an on-board computer (Intel® Core™ i7-1185G7E, 4.40 GHz) at a frequency of 200 Hz. The hydraulic joint servos are run in the underlying DSP at 1 kHz. The communication between the computer and underlying hardware is implemented under the ROS2-control framework. Considering the convergence speed and stability of the controller comprehensively, the prediction horizon of the MPC is set to $N_p = 35$, with the predictive sampling period of 0.01 s. The dynamic matrices and kinematic Jacobians are calculated using the open-source Pinocchio library [31]. All QP problems are solved using the operator splitting quadratic program solver [32] based on alternating direction method of multipliers.

The experiment will evaluate the velocity tracking performance of the robot under disturbances from unknown payloads and unknown terrain in the sagittal plane, corresponding to the two forms of disturbances in Eq. (12). Aiming to conduct a comprehensive assessment and distinguish the advantages of the proposed control method, three control strategies are implemented in each scenario: the LQR-based wheel torque regulation strategy [33], the MPC-based nominal CoM planning method [13], and the disturbance adaptive controller proposed in this paper.

4.1 Unknown payload experiments

The experiments in this section are designed to evaluate the stability and velocity tracking performance of the robot under unknown payload disturbances. Two sets of experiments were conducted: one focuses on assessing the response of the robot to abrupt payload changes, and the other focuses on velocity tracking under constant payloads.

In the abrupt payload change experiments, the reference forward velocity of the robot was set to 0 m/s; after which, a 50 N payload was applied to the end of the arm. Notably, the robot did not have prior precise information regarding the load magnitude. The performance of the proposed method is evaluated by comparing the response of the robot under the three previously mentioned control strategies. The responses under different control strategies are shown in Fig. 4. As presented in the experimental results in Fig. 5(a), the robot maintains stable motion under payload disturbances across all three control methods and eventually converges to a stable forward velocity. However, using the wheel torque regulation control strategy based on LQR, the steady-state error in the robot's speed tracking reached 0.6 m/s, while the error under the nominal MPC-based control method was 0.38 m/s. In contrast, integrating the compensation mechanism of the disturbance estimator, the disturbance-adaptive MPC demonstrates an impressive capability to gradually converge the robot's velocity to 0 m/s within 5 s, achieving an average steady-state velocity tracking error of 0.003 m/s. Figure 5(b) depicts the CoM trajectory of the robot. In the absence of payload disturbances, the robot's CoM is located at the equilibrium point directly above the wheels. However, under the payload disturbances, the robot adjusts its CoM backward to counterbalance the impact of the payload force on the system. Although the CoM position of the robot converges to the same equilibrium point across all three control modes, the adaptive MPC-based control method offers a faster adjustment speed, enabling the robot to achieve superior resistance to disturbances.

During a transportation task, the robot encounters continuous and constant payload disturbances. Thus, accurate velocity tracking is crucial for enhancing the



(a) Adaptive MPC proposed in this work (b) LQR based torque regulation

Fig. 4 Screenshots of robot responses to unknown abrupt payload changes under the (a) adaptive MPC-based control method and the (b) LQR-based wheel torque regulation strategy.

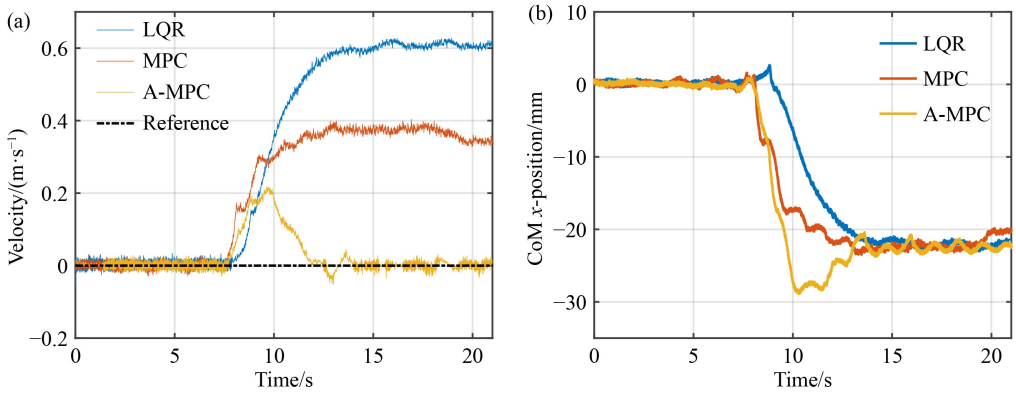


Fig. 5 Experimental results of the robot's response under unknown abrupt payload changes. (a) Forward velocity response of the robot. (b) CoM position response trajectories. The blue line represents the LQR-based wheel torque adjustment method, the red line corresponds to the nominal MPC method, and the yellow line shows the adaptive MPC-based control method proposed in this paper. The black dotted line indicates the reference forward velocity, which is set to 0 in this experiment. The x-direction corresponding to the forward moving direction. A-MPC: adaptive MPC.

precision of the robot's task execution. Aiming to evaluate its accuracy, a set of velocity tracking experiments were conducted under a constant payload, as illustrated in Fig. 6. Initially, the reference velocity of the robot was set to 0 m/s, and a constant payload of 50 N was applied at its elbow joint. After the velocity of the robot stabilized, a step signal was issued, adjusting the reference velocity from 0 to 0.5 m/s and back to 0 m/s. The velocity tracking performance of the robot under three control methods was then compared.

The experimental results in Fig. 7 indicate that when the reference velocity was 0 m/s, the velocity tracking steady-state error was 0.4 and 0.2 m/s under the LQR-based wheel torque regulation strategy and the nominal MPC method, respectively. However, the adaptive MPC-based control method proposed in this paper could reduce the error to within 0.02 m/s. When the reference velocity changed, the wheeled biped robot exhibited a velocity undershoot at the start of acceleration due to its inherent non-minimum

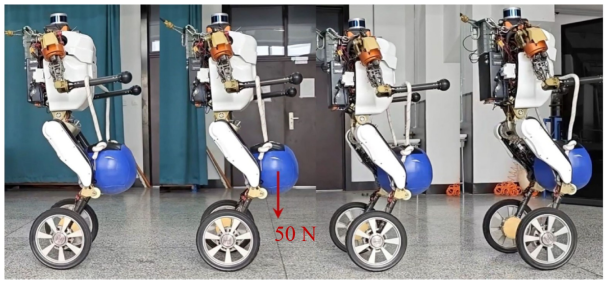


Fig. 6 Screenshot of the experimental process under constant and continuous unknown payload disturbances. The 50 N payload was applied around the robot's elbow joints.

phase characteristics, thereby delaying the velocity response. Despite this delay, under the adaptive MPC regulation, the velocity of the robot converged to the reference value with an average steady-state tracking error within 0.01 m/s. In contrast, the other two control methods showed substantial errors and delays in the velocity response of robots, which potentially

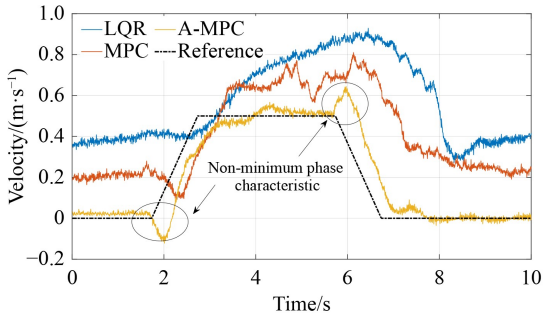


Fig. 7 Velocity tracking experimental results of WLR-IV under constant and continuous unknown payload disturbances with the regulation of three control methods. A-MPC: adaptive MPC.

compromises trajectory tracking performance during autonomous navigation. These results demonstrate that the control method proposed in this paper effectively addresses constant and continuous unknown payload disturbances while still maintaining relatively high velocity tracking accuracy, thereby enhancing the robot's performance in autonomous transportation tasks.

4.2 Unknown terrain experiments

For wheeled robots, changes in ground slope can cause variations in the wheel–ground contact force, introducing force disturbances into the nominal model of the robot. Aiming to evaluate the robustness and adaptability of the proposed method in this paper under unknown ground slope disturbances, a series of velocity tracking experiments traversing over uneven terrain were conducted. The experimental process is illustrated in Fig. 8. The robot moved at a reference velocity of 0.5 m/s on a slope with an inclination of 8.2°, and the velocity tracking performance was compared across three control methods. The experimental results are shown in Fig. 9. The robot began ascending the slope at approximately 2.2 s. During its ascent, the forward velocity of the robot substantially decreased under all three control methods. However, with the regulation of adaptive MPC-based control method, the maximum velocity tracking error was limited within 0.09 m/s. This error is considerably smaller than the 0.3 and 0.18 m/s errors observed with the LQR-based wheel torque adjustment strategy and the nominal MPC method, respectively. During the subsequent descent, the adaptive MPC-based control method maintained the velocity tracking error within 0.1 m/s, outperforming the other two control methods, which had errors of 0.5 and 0.12 m/s. Upon returning to flat ground, all three methods eventually converged to the reference forward velocity, with the adaptive MPC achieving faster convergence. These experimental results

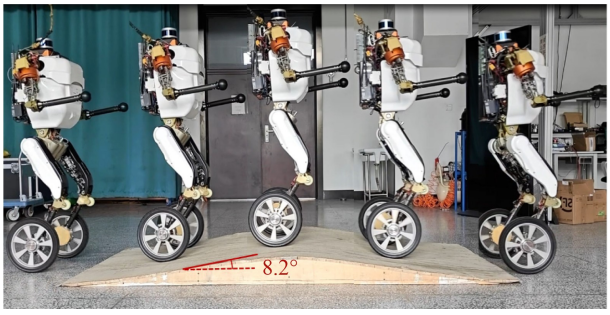


Fig. 8 Screenshot of the robot velocity tracking experiment on unknown slope terrain with an inclination of 8.2°. The reference velocity was set as 0.5 m/s.

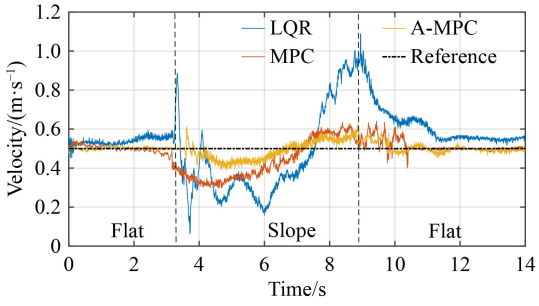


Fig. 9 Velocity tracking experimental results of WLR-IV under the disturbance of unknown slope terrain with the regulation of three control methods. A-MPC: adaptive MPC.

demonstrate that the proposed method allows the robot to adapt to disturbances introduced by uneven terrains while maintaining accurate velocity tracking.

Figure 10 compares the average velocity tracking errors and their standard deviation for each control strategy across three sets of experiments. Notably, under the regulation of the adaptive MPC, the

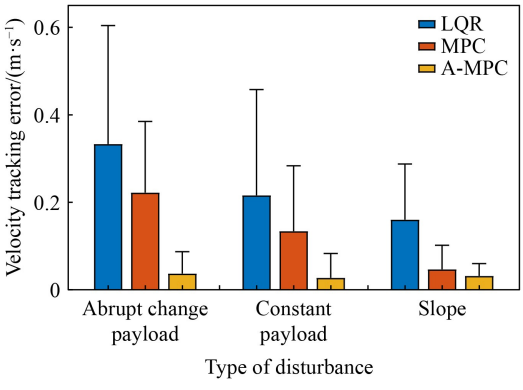


Fig. 10 Comparison of the average velocity tracking errors and their standard deviation among three control strategies across three types of disturbances. In the statistical analysis, error values are taken as absolute values to effectively reflect the overall tracking accuracy.

robot's velocity tracking error and its standard deviation across the three types of disturbances are substantially reduced compared to the other two methods. This finding intuitively demonstrates that the proposed disturbance-adaptive MPC method is highly effective at resisting the effect of uncertain disturbances on the robot system, substantially improving the forward velocity tracking accuracy of the robot.

5 Conclusions and future work

This paper proposes a disturbance-adaptive control framework for wheeled biped robots based on Kalman filtering and MPC. The method uses the inherent dynamic coupling characteristics of wheeled biped robots, utilizing only the data from robot's IMU and encoders. This framework enables the online estimation of the comprehensive disturbance effects on the robot in the sagittal plane, including unknown payloads, uneven terrains, and inertia model errors, without requiring additional force/torque sensors. These disturbances are then compensated for by the nominal MPC framework to obtain a highly accurate predictive model, allowing the accurate regulation of the forward velocity by planning the CoM dynamic trajectory of the robot. Finally, a QP-based whole-body controller maps the CoM dynamics into joint space, generating the robot's full-body motion trajectory, which achieves stable motion under uncertain disturbance. Comparing velocity tracking performance under various disturbances with other two widely used control methods in wheeled biped robots, the experimental results demonstrate that the disturbance-adaptive MPC method proposed in this work can effectively reduce the impact of uncertain disturbances on velocity tracking. This reduction allows the robot to maintain stable motion and achieve high velocity tracking accuracy, providing a foundation for precise autonomous mobility and operational tasks. However, the proposed method uses WLIP as the predictive model for MPC, which is based on the constant CoM height constraint. During typical robot motions, the variation in CoM height is limited, and the reference CoM height is updated at each control node, enabling the method to ensure velocity tracking accuracy in most motion scenarios. In the cases where the robot performs high-maneuverability movements involving rapid changes in CoM height, relaxing the CoM height constraint and utilizing a nonlinear variable-length WIP as the predictive model are necessary. These conditions represent a limitation of the proposed method. However, the robot's balance is maintained by the WBC; thus, the modeling errors associated with the WLIP do not compromise the stability of the system.

Future work intends to integrate visual sensors and LiDAR, aiming to further enhance the accuracy of robot state estimation and environmental perception. Through the integration of the disturbance estimation and adaptive control method into an autonomous wheeled biped robot system, the authors aim to endow robots with autonomous mobility and precise operation in complex and unstructured environments.

Nomenclature

Abbreviations

CoM	Center of mass
DoF	Degree of freedom
F/T	Force/torque
GRF	Ground reaction force
IMU	Inertial measurement unit
LiDAR	Light detection and ranging
LQR	Linear quadratic regulator
MPC	Model predictive control
QP	Quadratic programming
WBC	Whole-body control
WIP	Wheeled inverted pendulum
WLIP	Wheeled linear inverted pendulum

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